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Chapter 9

Data Levels, Color Management, and ACES

This chapter covers operational details that affect how color is managed for media that is imported into and exported from DaVinci Resolve. If color accuracy is important to you, then it's a good idea to learn more about how Resolve handles the data levels of each clip, how DaVinci Resolve Color Management helps you to work with different formats, and how to use ACES.

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Data Levels Settings and Conversions

Different media formats use different ranges of values to represent image data. Since these data formats often correspond to different output workflows (cinema vs. broadcast), it helps to know where your project's media files are coming from, and where they're going, in order to define the various data range settings in DaVinci Resolve and preserve your program's data integrity.

To generalize, with 10-bit image values (with a numeric range of 0–1023), there are two different data levels (or ranges) that can be used to store image data when writing to media file formats such as QuickTime, MXF, or DPX. These ranges are:

- **Video:** Typically used by Y'CBCR video data. All image data from 0 to 100 percent must fit into the numeric range of 64–940. Specifically, the Y' component's range is 64–940, while the numeric range of the CB and CR components is 64–960. The lower range of 4–63 is reserved for "blacker-than-black," and the higher ranges of 941/961–1019 are reserved for "super-white." These "out of bounds" ranges are recorded in source media as undershoots and overshoots, but they're not acceptable for broadcast output.

- **Full:** Typical for RGB 444 data acquired from digital cinema cameras, or film scanned to DPX image sequences. All image data from 0 to 100 percent is simply fit into the full numeric range of 4 to 1023.

Keep in mind that every digital image, no matter what its format, has absolute minimum and maximum levels, referred to in this section as 0–100 percent. Whenever media using one data range is converted into another data range, each color component's minimum and maximum data levels are remapped so that the old minimum value is scaled to the new data level minimum, and the old maximum value is scaled to the new data level maximum;

- (minimum Video Level) 64 = 4 (Data Level minimum)
- (maximum Video Level) 940 or 960 = 1023 (Data Level maximum)

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Converting Between Ranges and Clipping

Simply converting an image from one data range to another should result in a seamless change. All "legal" data from 0–100 percent is always preserved and is linearly scaled from the previous data range to fit into the new data range.

The exceptions to this are undershoots and overshoots that you've deliberately set, also referred to as out-of-bounds levels. The overshoots and undershoots that are allowable in "Video Levels" media (known as sub-black or super-black and super-white) are usually clipped when converted to full-range "Full Levels." However, DaVinci Resolve preserves this data internally, and these clipped pixels of detail in the undershoots and overshoots are still retrievable by making suitable adjustments in the Color page to bring them back into the "legal" range.

The out-of-bounds image data that's preserved within the headroom of Video Levels by DaVinci Resolve while working is usually clipped, however, when you either output to video or render your output. There are two settings that let you get around this for instances where you want to preserve these levels:

A checkbox in the Video Monitoring group of the Master settings, "Retain sub-black and super-white data," lets DaVinci Resolve output undershoots (sub-black) and overshoots (super-white) to video when Data Level is set to Video. When this is turned off, these out-of-bounds values are clipped on output.

A checkbox in the Advanced settings of the Render settings in the Deliver page, "Retain sub-black and super-white data," lets DaVinci Resolve render undershoots (sub-black) and overshoots (super-white) to exported media when Data Level is set to Video.

Internal Image Processing and Clip Data Levels

It's useful to know that, internally to DaVinci Resolve, all image data is processed as full range, uncompressed, 32-bit floating point data. What this means is that each clip in the Media Pool, whatever its original bit-depth or data range, is scaled into full-range 32-bit data. How each clip is scaled depends on its Levels setting in the Clip Attributes window, available from the Media Pool contextual menu.

Selecting Auto, Video, or Full levels

By converting all clips to uncompressed, full-range, 32-bit floating point data, Resolve guarantees the highest quality image processing that's possible. As always, the quality of your output is dependent on the quality of the source media you're using, but you can be sure that Resolve is preserving all the data that was present in the original media.

Assigning Clip Levels in the Media Pool

When you first import media into the Media Pool, either manually in the Media page or automatically by importing an AAF or XML project in the Edit page, Resolve automatically assigns the "Auto" Levels setting. When a clip is set to Auto, the Levels setting used is determined based on the codec of the source media.

DaVinci Resolve generally does a good job of figuring out the appropriate Levels setting of each clip on its own. However, in certain circumstances, such as when you're working with media that was originated in one format but transcoded into another, you may find that you need to manually choose the appropriate settings so that the levels of each clip are interpreted correctly. This can be done using each clip's Levels setting in the Clip Attributes window, available from the Media Pool contextual menu in either the Media or Edit pages.

To change a clip's Data Level setting:

1. Open the Media or Edit page.
2. Select one or more clips, then right-click one of them and choose Clip Attributes.
3. Click the Levels ratio button corresponding to the data level setting you want to assign, then click OK.

TIP: If you need to change the Levels setting of a range of clips that share a unique property such as reel name, resolution, frame rate, or file path, you can view the Media Pool by column, and sort by the particular column that will best isolate the range of media to which you need to make a data level assignment.

Once you change a clip's Levels setting, that clip will automatically be reconverted based on the new assignment. If it appears to be correct, then you're ready to work. If it doesn't, then you may want to reconsider the Levels assignment you've made, and you should check with the person who provided the media to find out how it was generated, captured, and exported.

So long as the Levels settings used by your clips are accurate, you should be ready to work. However, problems can still occur based on what external video hardware you're using with your workstation, and how you need to deliver the finished media to your client. For this reason, there are three additional data level settings that you can use to maintain data integrity, while at the same time seeing the proper image as you work.

Video Monitoring Data Levels

Superficial problems may result if the settings used by your external display differ from the settings you're using to process data levels in Resolve. Accordingly, there is a Video/Full Level setting in the Master Settings panel of the Project Settings (in the Video Monitoring section).

When you change this setting, the image being output to your external display should change, but the image you see in your Viewer will not. That's because this setting only affects the data levels being output via the video interface connecting the Resolve workstation to your external display. It has no effect on the data that's processed internally by Resolve, or on the files written when you render in the Deliver page.

There are two options:

Video: This is the correct option to use when using a broadcast display set to the Rec. 709 video standard (10-bit 64–940).

Full: If your monitor or projector is capable of displaying "full range" video signals, and you wish to monitor the full 10-bit data range (4–1023) while you work, then this is the correct option to use.

It is imperative that the option you choose in DaVinci Resolve matches the data range the external display is set to. Otherwise, the video signal will appear to be incorrect, even though the internal data is being processed accurately by DaVinci Resolve.

Auto/Video/Full Level selection for monitoring

Deck Capture and Playback Data Level

There is a separate "Video/Data Level" setting that is specific to when you're capturing from or outputting to VTRs. This setting also affects the video signal that is output via the video interface connecting the Resolve workstation to your VTR (which is usually also in the signal chain used for monitoring). However, it only takes effect when you're capturing from tape in the Media page, or editing to tape in the Deliver page. If you never capture or output to tape, this setting will never take effect.

This setting is found in the Deck Capture and Playback panel of the Project Settings.

The reason for a separate option for tape capture and output is that often you'd want to monitor in one format (normally scaled Rec. 709), but output to tape in another (full range RGB 444). This way, you can set up Resolve to accommodate this workflow, and then not have to worry about manually switching your video interface back and forth.

There are two options:

Video: This is the correct option to use when you want to output conventional Rec. 709 video to a compatible tape format.

Full: This is the correct option to use when you want to output "full range" RGB 444 video to a compatible tape format.

Once tape ingest or output has finished, your video interface goes back to outputting using the setting specified by the "Colorspace conversion uses" setting in the Master Settings panel of the Project Settings (in the Video Monitoring section).

Output Data Level Settings in the Deliver Page

Finally, there's one last set of data level settings, available in the Render Settings list, within the Format group. It's the "Set to video or data level" drop-down menu. It's there to give you the ability to convert the data level of your rendered output, if necessary.

All media is output using a single data level, depending on your selection. There are three options:

Automatic: The output data level of all clips is set automatically based on the codec you select to render to in the "Render to" drop-down menu.

Video: All clips are rendered as normally scaled for video (10-bit 64–940).

Full: All clips are rendered as full range (10-bit 4–1019).

For most projects, leaving this setting on "Automatic" will yield the appropriate results. However, if you're rendering media for use by another image processing application (such as a compositing application) that is capable of handling "full range" data, then full range output is preferable for media exchange as it provides the greatest data fidelity. For example, when outputting media for VFX work as a DPX image sequence, or as a ProRes 4444 encoded QuickTime file, choosing "Unscaled full range data" guarantees the maximum available image quality. However, it is essential that the application you use to process this media is set to read it as "full range" data, otherwise the images will not look correct.

So, What's the "Proper" Data Range for Output?

Strictly speaking, there is no absolutely "proper" data range to use when outputting image data. As long as the Levels setting of each clip in the Media Pool is set to reflect how each clip was created, your primary consideration is which data range is compatible with the media format or application you're delivering to. If the media format you're exporting to supports either normally scaled or full range, and the application that media will be imported into supports either normally scaled or full range, then it's really your choice, as long as everyone involved with the project understands how the data range of the media is meant to be interpreted once they receive it.

Outputting to hardware is a bit trickier, in that you need to make sure that the external display or VTR you're outputting to is set up to receive a signal using the data range you've chosen. If the device is limited to only one data range, then you need to be sure that you're outputting to it using that data range, or the levels of the image will appear to be incorrect, even though the image data being processed by Resolve is actually fine.

Introduction to DaVinci Resolve Color Management

How color is managed in DaVinci Resolve depends on the "Color Science" setting at the top of the Color Management panel of the Project Settings. There are four options: DaVinci YRGB, DaVinci YRGB Color Managed, DaVinci ACEScc, and DaVinci ACEScct. This section discusses the second setting, DaVinci YRGB Color Managed. ACEScc and ACEScct is discussed in the following section in this chapter.

Display Referred vs. Scene Referred Color Management

The default DaVinci YRGB color science setting, which is what DaVinci Resolve has always used, relies on what is called "Display Referred" color management. This means that Resolve has no information about how the source media used in the Timeline is supposed to look; you can only judge color accuracy via the calibrated broadcast display you're outputting to. Essentially, you are the color management, in conjunction with a trustworthy broadcast display that's been calibrated to ensure accuracy.

DaVinci Resolve 12 introduced a color science option called "DaVinci YRGB Color Managed," or more simply "Resolve Color Management" (RCM). This introduced a so-called "Scene Referred" color management scheme, in which you have the option of matching each type of media you've imported into your project with a color profile that informs DaVinci Resolve how to represent each specific color from each clip's native color space within the common working color space of the timeline in which you're editing, grading, and finishing.

This is important, because two clips that contain the same RGB value for a given pixel may in actuality be representing different colors at that pixel, depending on the color space that was originally associated with each captured clip. This is the case when you compare raw clips shot with different cameras made by different manufacturers, and it's especially true if you compare clips recorded using the differing log-encoded color spaces that are unique to each camera.

This Scene Referred component of color management via RCM doesn't do your grading for you, but it does try to ensure that the color and contrast from each different media format you've imported into your project are represented accurately in your timeline. For example, if you use two different manufacturer's cameras to shoot green trees, recording Blackmagic Film color space on one, and recording to the Sony SGamut3.Cine/SLog3 color space on the other, you can now use RCM to make sure that the green of the trees in one set of clips match the green of the trees in the other, within the shared color space of the Timeline.

It should be mentioned that this sort of thing can also be done manually in a more conventional Display Referred workflow, by assigning LUTs that are specific to each type of media, or using Color Space Transform Resolve FX in order to transform each clip from the source color space to the destination color space that you require. However, RCM's automation can make this process faster by freeing you from the need to locate and maintain a large number of LUTs to accommodate your various workflows. Also, the matrix math used by RCM (as well as the Color Space Transform operation) extracts high-precision, wide-latitude image data from each supported camera format, preserving high-quality image data from acquisition, through editing, color grading, and output. These are all advantages when compared to lookup tables, which can have plenty of precision, but can clip out-of-bounds image data and introduce issues when differing lookup table interpolation methods cause minor inconsistencies with color space transformations from application to application.

The preservation of wide-latitude image data deserves elaboration. LUTs clip image detail that goes outside of the numeric range they're designed to handle, so this often requires the colorist to make a pre-LUT adjustment to "pull back" image data in the highlights that you want to retrieve. Using RCM eliminates this two-step process, since the input color space matrix operations used to transform the source preserves all wide-latitude image data, making highlights easily retrievable without any extra steps.

Updated RCM In DaVinci Resolve 17

In version 17, DaVinci Resolve introduced the biggest improvements to Resolve Color Management (RCM) since it was originally introduced, adding numerous features to simplify setup, improve image quality, and make the "feel" of your grading controls more consistent. Specific improvements include improved metadata management for incoming media files that support color metadata, a new wide gamut color space suitable for using as your default Timeline working color space for any program, a new Input Tone Mapping option (Input DRT) that makes it easier to mix media formats for SDR and HDR grading, improved Timeline to Output Tone Mapping (Output DRT) that offers improved shadow and highlight handling, and select color space-aware grading palettes that make controls feel and perform well no matter what you're grading.

This updated Resolve Color Management has the same name as the previous version. However, older projects using the previous version of RCM will have Color science set to Legacy, to preserve the older color management settings and color transformations effect on your work. For more information on how the previous generation of RCM works, see the September 2020 version of the DaVinci Resolve 16 Manual.

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How Is DaVinci Resolve Color Management Different from ACES?

This is a common question, but the answer is pretty simple. Resolve Color Management (RCM) and ACES are both Scene Referred color management schemes designed to solve the same problem. However, if you're not in a specific ACES-driven cinema workflow, DaVinci Resolve Color Management can be simpler to use, and will give you all of the benefits of color management, while approximating the "feel" that the DaVinci Resolve Color page controls have always had.

Resolve Color Management for Editors

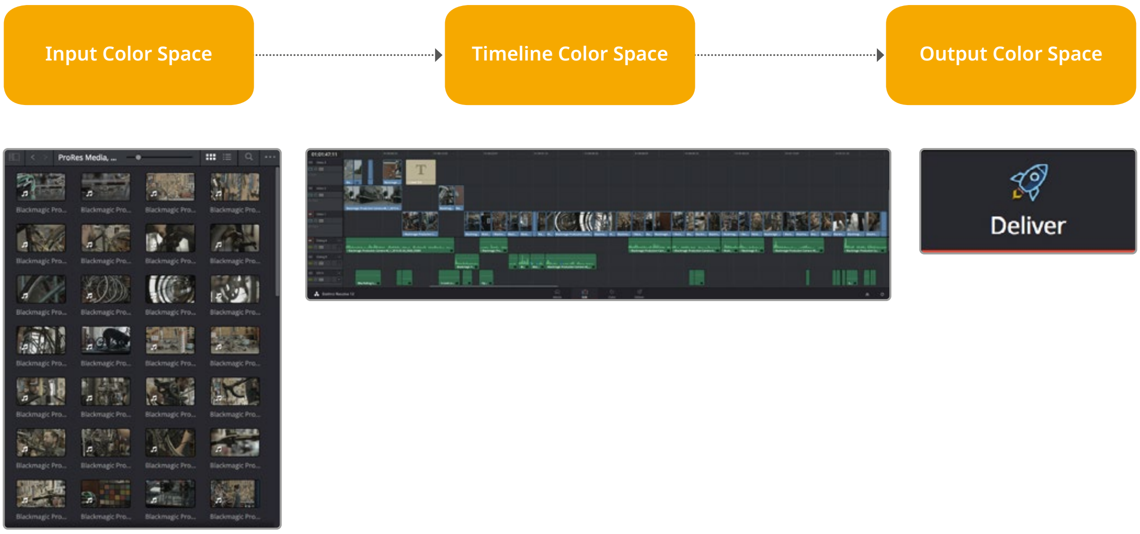
RCM isn't just for Colorists. RCM can be easier for editors to use in situations where the source material is log-encoded. Log-encoded media preserves highlight and shadow detail, which is great for grading and finishing, but it looks flat and unpleasant, which is terrible for editing.

Even if you have no idea how to do color correction, it's simple to turn RCM on in the Color Management panel of the Project Settings, and then use the Media Pool to assign the particular Input Color Space that corresponds to the source clips from each camera. Once that's done, each log-encoded clip is automatically normalized to the default Timeline Color Space of Rec. 709 Gamma 2.4. So, without even having to open the Color page, editors can be working with pleasantly normalized clips in the Edit page.

The Input, Timeline, and Output Color Space

The foundation of Resolve Color Management rests on three core settings. Not only do you have the ability to either automatically or manually identify the color science of each individual source clip (the Input Color Space), but you also have explicit control over the working color space within which all color adjustments and operations are made (the Timeline Color Space), and you have separate control over the Output Color Space that defines how your graded image will be monitored and output.

This means that, basically, Resolve Color Management consists of two color transforms working together, converting each source clip via its Input Color Space definition into the Timeline Color Space in which you work, and then converting the adjusted image from the Timeline Color Space to whatever Output Color Space you require to deliver the project.



Resolve Color Management consists of three color transforms working together.

This means that, as a colorist, you can set the Timeline Color Space that you're working in to whatever you prefer. If you prefer grading wide-gamut log media because you like the way the grading controls behave in that color space, you can set the Timeline Color Space in the Color Management panel of the Project Settings to DaVinci Wide Gamut (more on this below), or any of the available log formats, including ARRI Log C, REDWideGamutRGB/Log3G10, and Cineon Film Log. If you instead prefer grading in the Rec. 709 color space because you're mastering a standard dynamic range (SDR) program to Rec. 709 and you're more comfortable with how the controls in DaVinci Resolve have always felt in that color space, you can choose that instead. Whatever Timeline Color Space you assign is what all source clips will be transformed to for purposes of making grading adjustments in the Color page, so you can make this choice using a single setting.

A key benefit of the color space conversions that RCM applies is that no image data is ever clipped during the Input to Timeline color space conversion. For example, even if your source is log-encoded or in a camera raw format, grading with a Rec. 709 Timeline Color Space does nothing to clip or otherwise limit the image data available to the RCM image processing pipeline. All image values greater than 1.0 or less than 0.0 are preserved and made available to the next stage of RCM processing, the Timeline to Output color space conversion.

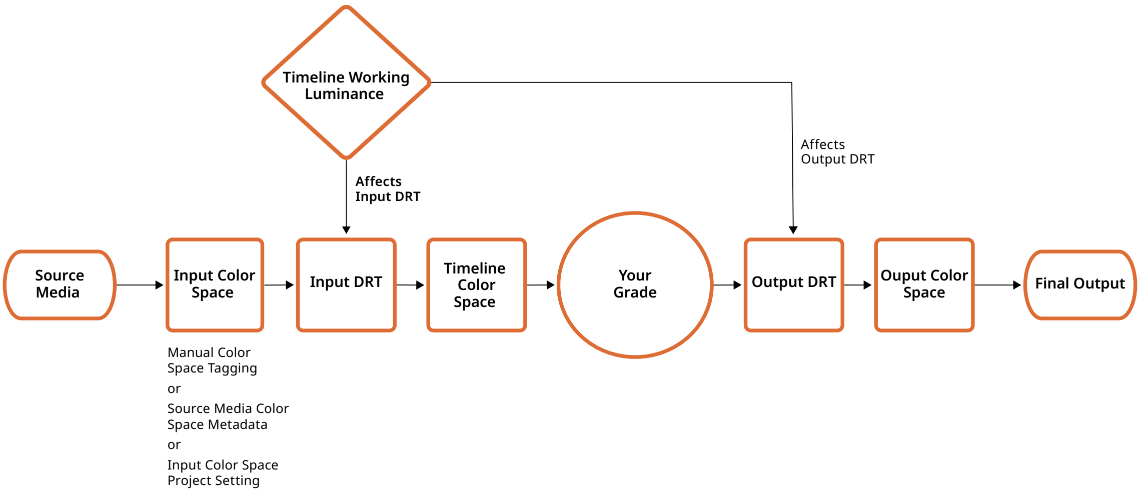
Consequently, if you're grading in a color space other than the one you need to output to, you don't have to worry about data loss during the color transformation back to the color space you actually want to output to. The Output Color Space setting gives you the freedom to work using whatever Timeline Color Space you like while grading, with Resolve automatically converting your output to the specific color space you want to monitor with and deliver to. And thanks to the precision of the image processing in DaVinci Resolve, you can convert from a larger color space to a smaller one and back again without clipping or a loss of quality. Of course, if you apply a LUT or use Soft Clip within a grade, then clipping will occur, but that's a consequence of using those particular operations.

TIP: If you want to use Resolve Color Management, but you want the Input and Output Color Spaces to match whatever you set the Timeline Color Space to, you can choose "Bypass" in the "Input Colorspace" and "Output Colorspace" drop-down menus.

Finally, it is the Output Color Space that determines the final color space of your rendered result. While no image data is clipped during the Source to Timeline color space conversion, image data will be clipped during the Timeline to Output color space conversion in order for the final image to conform to the color space being rendered and output, unless you use the Gamut Mapping options to compress image data during the Timeline to Output Color Space conversion.

The RCM Image Processing Pipeline

The previous explanation is, of course, simplified. To clarify the inner workings of Resolve Color Management for advanced users, the following flowchart presents a rudimentary overview of how every parameter works together to automatically manage the color of clips in your program.



Resolve Color Management's image processing pipeline, illustrated

Identifying the Input Color Space of Different Clips

Central to the process of automated color management is knowing the color space and transfer function used by every clip of source media in your project. There are a variety of ways DaVinci Resolve can figure this out, in a cascading decision-tree that can be manually overridden if necessary. Deriving the Input Color Space involved the following stages of automated decision making:

1. If the source media is a camera raw format like .braw, .R3D, .ari, etc., DaVinci Resolve uses manufacturer-supplied colorimetry to automatically debayer the clip and identify its Input Color Space.
2. Otherwise, if the source media has embedded color space metadata (QuickTime or .MXF make this possible), then use that to identify the Input Color Space.
3. Otherwise, if there is no embedded color space metadata, use the default Input Color Space setting of the Project Settings to assign an Input Color Space to all otherwise unidentified clips.
4. If necessary, you can manually set the Input Color Space of clips in the Media Pool, which overrides both embedded color space metadata (in case it's wrong), or the default Input Color Space setting (if you're dealing with multiple color spaces). You cannot override the Input Color Space of camera raw media.

The following sections discuss each of these steps in more detail.

Using Camera Raw Formats

When you use RCM in a project that uses Camera Raw formats, color science data from each camera manufacturer is used to debayer each camera raw file to specific color primaries with linear gamma, so that all image data from the source is preserved and made available to DaVinci Resolve's color managed image processing pipeline. As a result, the Camera Raw project settings and Camera Raw palette of the Color page are disabled, because RCM now controls the debayering of all camera raw clips, and all image data from the raw file is available no matter which Timeline Color Space you choose to work within.

Using Source Media Color Space Metadata

When enabled, RCM automatically identifies the color space information of imported media that's been either transcoded or recorded directly to supported non-raw media formats, reading the NCLC metadata of QuickTime-wrapped files, the color space metadata of .mxf-wrapped files, and the XML sidecar files that track color management in ACES workflows. This behavior is automatic; there are no visible controls governing this behavior aside from the individual Input Color Space and Input Gamma settings associated with each clip in the Media Pool.

Color Space Metadata in QuickTime

DaVinci Resolve is capable of reading the NCLC metadata found within media files wrapped within a QuickTime container for proper color management. This metadata consists of three values formatted as (for example) 1-1-1. From left to right, these three digits specify the Color Primary (or color space), Transfer Function (or gamma), and Color Matrix used by that media file.

These values are standardized in the SMPTE Registered Disclosure Document RDD 36:2015. For your information, the different codes are listed in the following table. In the previous example, the code of 1-1-1 indicates a standard dynamic range clip that uses the BT.709 primaries, transfer function, and color matrix.

Color Primary		Transfer Function		Color Matrix	
0	Reserved	0	Reserved	0	GBR
1	ITU-R BT.709	1	ITU-R BT.709	1	BT709
2	Unspecified	2	Unspecified	2	Unspecified
3	Reserved	3	Reserved	3	Reserved
4	ITU-R BT.470M	4	Gamma 2.2 curve	4	FCC
5	ITU-R BT.470BG	5	Gamma 2.8 curve	5	BT470BG
6	SMPTE 170M	6	SMPTE 170M	6	SMPTE 170M
7	SMPTE 240M	7	SMPTE 240M	7	SMPTE 240M
8	FILM	8	Linear	8	YCOCG
9	ITU-R BT.2020	9	Log	9	BT2020 Non-constant Luminance
10	SMPTE ST 428-1	10	Log Sqrt	10	BT2020 Constant Luminance

Color Primary		Transfer Function		Color Matrix	
11	DCI P3	11	IEC 61966-2-4	–	–
12	P3 D65	12	ITU-R BT.1361 Extended Colour Gamut	–	–
–	–	13	IEC 61966-2-1	–	–
–	–	14	ITU-R BT.2020 10 bit	–	–
–	–	15	ITU-R BT.2020 12 bit	–	–
–	–	16	SMPTE ST 2084 (PQ)	–	–
–	–	17	SMPTE ST 428-1	–	–
–	–	18	ARIB STD-B67 (HLG)	–	–

The Default Input Color Space

The default Input Color Space can only be set if the "Resolve color management preset" drop-down menu is set to Custom. Otherwise, it defaults to "Rec. 709 Gamma 2.4" for all presets. Or else, this setting is the default color space that all otherwise unidentified clips in the Media Pool will default to.

Manually Tagging Clip Color Space

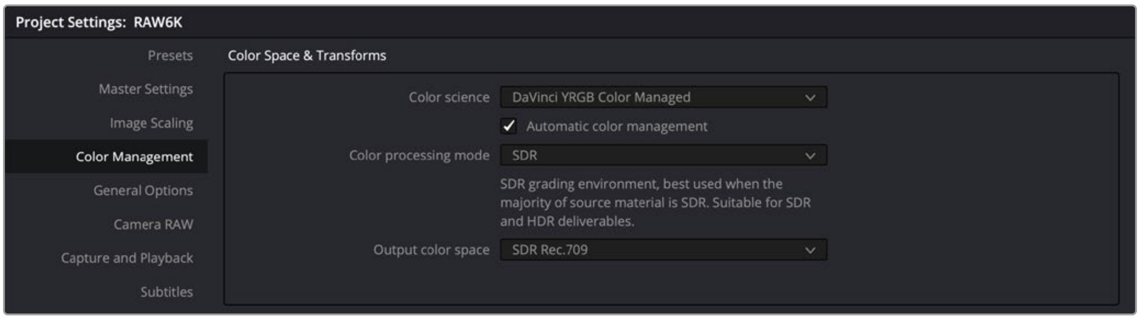
If necessary, you can manually identify the color space of one or more selected clips in the Media Pool by right-clicking them and choosing the Input Color Space (and optionally the Input Gamma) from the contextual menu.

Simple RCM Setup

When you first choose DaVinci YRGB Color Managed from the Color science drop-down menu of the Color Management panel in the Project Settings, you're presented with a simple pair of menus for setting up how you want to work with Resolve Color Management: the "Resolve color management preset," and the "Output Color Space."

Automatic Color Management

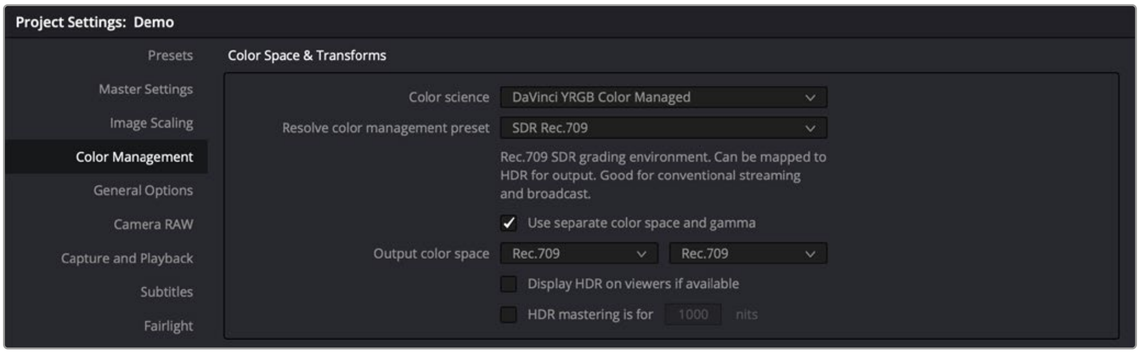
The first option when using RCM is to decide to use either Automatic Color Management or the Manual Presets. When the Automatic Color Management box is checked, DaVinci Resolve presents you with a simplified set of options for the most common use cases. For the Color Processing Mode, you choose SDR or HDR, and based on the file types and codecs in the Media Pool, DaVinci Resolve will automatically choose the appropriate input color space. Then, select from a list of common Output color spaces for delivery. If you want specific control of these parameters, uncheck Automatic Color Management box and select from the Color Management Presets below.



Automatic Color Management presets for fast, simple color management set up

Resolve Color Management Presets

The Resolve Color Management preset menu lets you choose how you want to use RCM to grade your program. Each of these presets fully configures your project's use of color management, and the setting you select directly impacts how you'll grade your program. Because of this, once you choose a method of working and you grade every clip in your program, those grades rely on the preset you used being selected in order to appear as they should.



Resolve Color Management presets for manual color management set up

When it comes to choosing a preset, a good way to think about which to use is to choose an SDR or HDR preset that corresponds to the primary deliverable you plan on outputting. Both SDR and HDR presets have several variations that you can choose among.

While these presets correlate to how you plan on outputting your program, they don't lock you in, since you can always change the Output Color Space (described below). This makes it possible to export multiple versions of your program, each intended for different venues, no matter which color management preset you're using.

Whenever you choose a preset, a brief description explains the workflow that preset is intended to facilitate. Here's a list of the available presets, with slightly more detailed explanations.

- **SDR Rec.709:** (default) Sets up a Rec. 709 SDR grading environment. Your work can be converted to HDR on output, if specified, but is limited to a Rec. 709 gamut with out-of-bounds colors being clipped. Gamma 2.4 is not mentioned in the name because scene versus display OOTF is managed automatically. Suitable for conventional streaming and broadcast.
- **SDR P3 Broadcast:** Sets up a P3-D65 SDR grading environment. Your work can be mapped to HDR for output, if specified, but it is limited to a P3-D65 gamut with out-of-bounds colors being clipped. Gamma 2.4 is not mentioned in the name because scene versus display OOTF is managed automatically. Suitable for wider gamut streaming and broadcast at SDR levels.

- **SDR P3 Cinema:** Sets up a P3-D60 SDR grading environment. Your work can be mapped to HDR for output, if specified, but it is limited to a P3-D60 gamut with out-of-bounds colors being clipped. Suitable for conventional Cinema projection.
- **SDR Rec.2020:** Sets up a Rec. 2020 SDR grading environment. Your work can be mapped to HDR for output, if specified. Good for wide gamut streaming and broadcast.
- **DaVinci Wide Gamut:** Sets up an extra wide gamut grading environment that's suitable for grading either SDR or HDR. Capable of exporting with maximum image fidelity, preserving highlight details of up to 10,000 nits. This is a log-encoded grading space for colorists wishing to work that way. Suitable for creating mezzanine intermediates or final deliverables, or for grading HDR with high nit levels.
- **HDR P3 Broadcast:** Sets up a P3-D65 HDR grading environment. Output gamut is limited to P3-D65, with out-of-bounds colors being clipped. Suitable for grading wide gamut SDR or HDR up to 1000 nits.
- **HDR Rec.2020:** Sets up a Rec. 2020 HDR grading environment. Suitable for wide gamut SDR or HDR deliverables up to 1000 nits.
- **Custom:** If none of the available presets suits how you need to work, you can choose Custom, which exposes the full set of RCM settings for you to set up to suit your needs.

IMPORTANT: For all presets, importing media that's in an identical or smaller gamut maps the image data into the larger color space of the preset without transforming it. Importing media with a wider gamut than the color space of the preset remaps the image data to fit into the smaller color space, while preserving as much image detail as possible.

Output Color Space

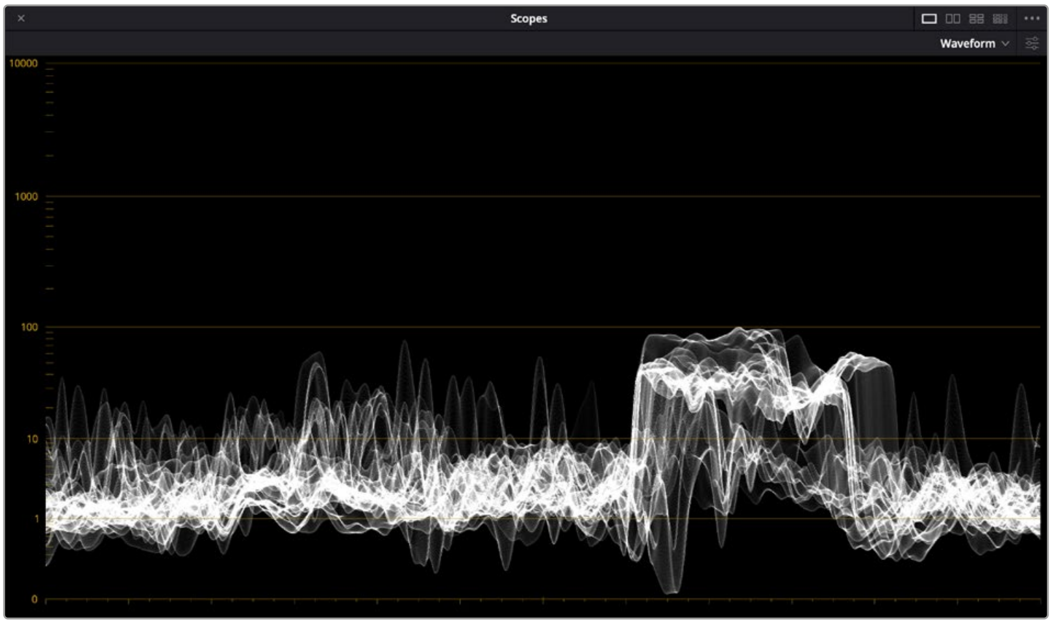
For most DaVinci Resolve installations and projects, you'll set your Output Color Space to match the needs of your program, according to your display's capabilities (or the capabilities your display is set to use for the project at hand). You'll also typically use a Resolve Color Management preset that matches those capabilities.

However, RCM gives you the flexibility of grading in one color space and then outputting to others, when necessary. For example, it's easy to grade an SDR Rec. 709 version of a program for streaming or broadcast, and then switch the Output Color Space to SDR P3 Cinema to output an additional deliverable for theatrical exhibition.

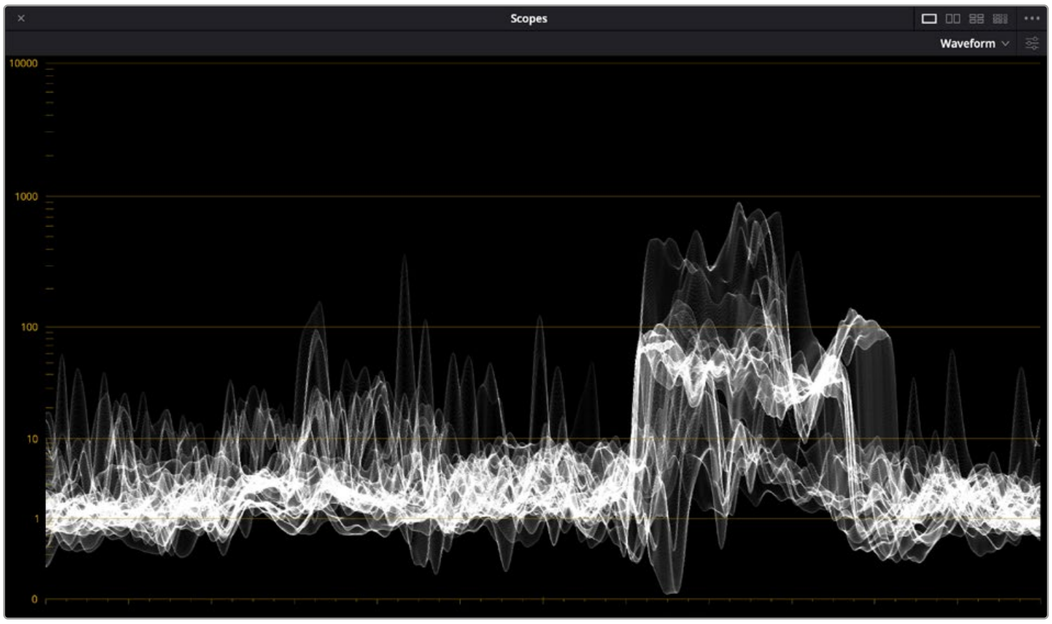
To facilitate this, you can set the Output Color Space to any setting, independent of the Resolve Color Management preset you've selected, and DaVinci Resolve will automatically convert from your Color Management Preset to the Output Color Space of your choice. When you do so, here are the rules that govern the resulting image transform.

When going SDR to HDR:

- 0-50 nits (18% mid-gray) in your program is mapped to 0-50 nits on output (no change).
- Everything from 51-90 nits in your program is remapped from 51 to 100 nits (slightly expanded).
- Everything from 91-100 nits in your program is remapped from 101 to 1000 nits (greatly expanded).



Original SDR grade seen within an HDR scale,



After an automatic SDR to HDR conversion

When going from HDR to SDR, the reverse is done:

- 0-50 nits (18% mid-gray) in your program is mapped to 0-50 nits on output (no change).
- Everything from 51 to 100 nits in your program is remapped from 51-90 nits (slightly compressed).
- Everything from 101 to 1000 nits in your program is remapped from 91-100 nits (greatly compressed).

MEDIA

While these methods of converting between SDR and HDR provide an effective starting point for conversion, they're not meant to be an automatic solution. It's critical that you do a trim pass whenever outputting a deliverable in a new color space and EOTF, so you can check every clip and make adjustments to improve the result when necessary.

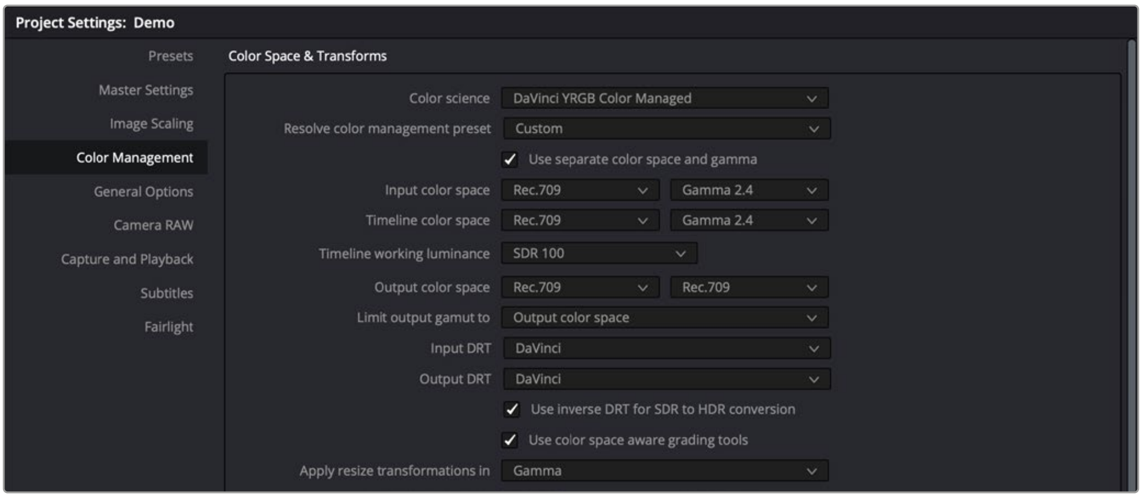
NOTE: When converting SDR to HDR, this behavior may exaggerate noise in imported SDR media that happens to have large flat expanses of bright colors. If you see particular clips that show this issue, you can disable this behavior on a clip by clip basis in the Media Pool clip contextual menu, or the Thumbnail Timeline contextual menu in the Color page, by toggling "Inverse DRT for SDR to HDR Conversion."

Advanced RCM Setup

Advanced users who need more detailed control over every aspect of RCM can choose Custom from the Resolve Color Management preset menu. This exposes every control that's available, which opens a world of workflow possibilities for advanced users and post production facilities.

Because each of the settings encompasses a significant amount of functionality, the following sections cover each particular parameter in detail.

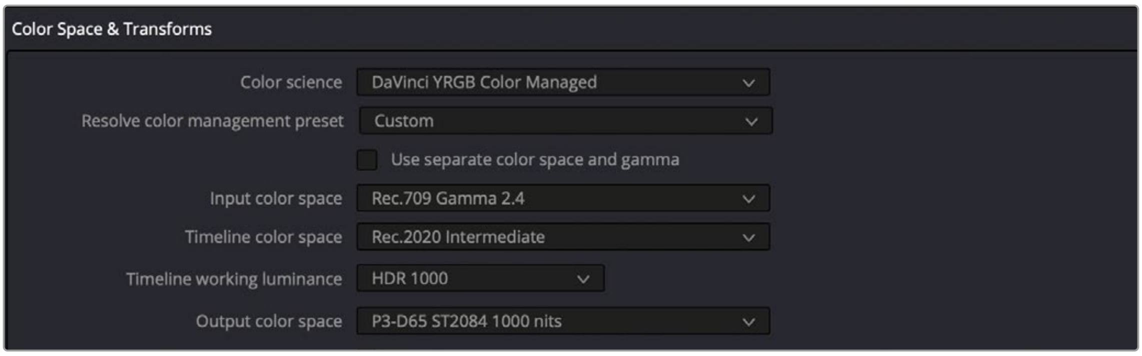
NOTE: Older projects using RCM will have Color science set to Legacy, to preserve the older color management settings and color transformations effect on your work. For more information on how the previous generation of RCM works, see the September 2020 version of the DaVinci Resolve 16 Manual.



Custom Color Management settings of Resolve Color Management, as updated in DaVinci Resolve 17

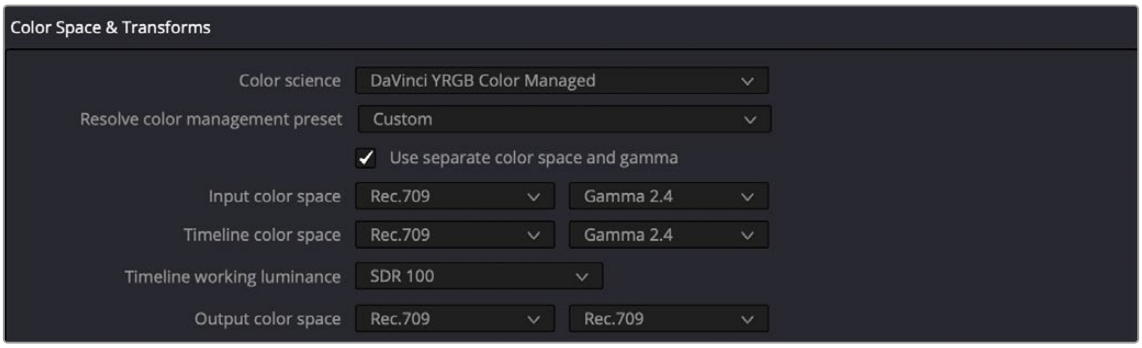
Single Setting vs. Dual Setting RCM

There are two ways you can set up RCM. When the "Use Separate Color Space and Gamma" checkbox is turned off, the Color Management panel of the Project Settings exposes one drop-down each for the Input, Timeline, and Output Color Space settings. Each setting lets you simultaneously transform the gamut and gamma, depending on which option you choose. This makes it a bit simpler to set up the transform you need.



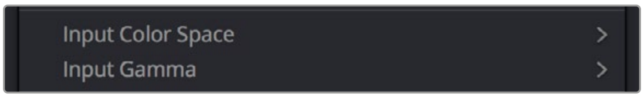
Single setting Resolve Color Management

If you turn the “Use Separate Color Space and Gamma” checkbox on, then the Color Management panel changes so that the Input, Timeline, and Output Color Space settings each display two pop-ups. The first drop-down lets you explicitly set the gamut, while the second drop-down lets you explicitly set the gamma. This makes it easier to see exactly which pair of transforms is being used at each stage of RCM.



Dual setting Resolve Color Management

Additionally, Dual Setting RCM enables you to assign separate gamut and gamma transforms to clips in the Media Pool.



Dual setting Resolve Color Management assignments for Media Pool clips

Setting the Input Color Space

This setting is the default color space that all otherwise unidentified clips in the Media Pool will default to, unless you manually identify the color space of these clips by right-clicking them and choosing an Input Color Space (and optionally Input Gamma) from the contextual menu.

This setting does not affect media in camera raw formats, or media with embedded color space metadata.

Choosing a Timeline Color Space

The Timeline Color Space is the "working" color space that determines how each clip's contrast and color are mapped for adjustment, which in turn has an impact on how sensitive the effects and grading controls are as you work. Some colorists prefer to work in the classic "video" color space of Rec. 709, since the controls feel comfortable and familiar, particularly if you're mastering SDR content. On the other hand, colorists who are used to working with log-encoded media (likely using the Log controls) often prefer to work in a more film-oriented workflow using Cineon, LogC, or other wide gamut, logarithmically encoded formats.

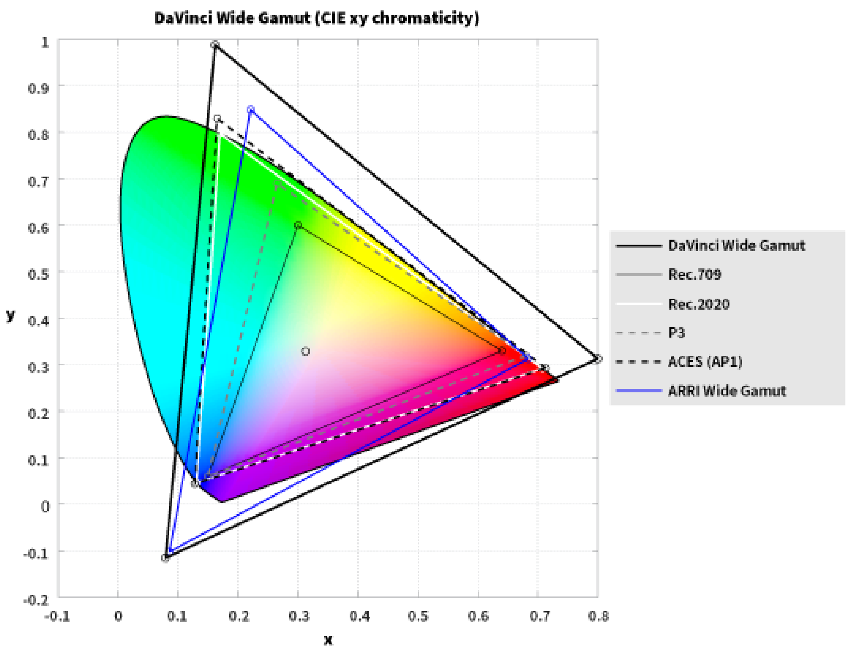
If you're outputting an SDR deliverable, any color space that you're comfortable will produce good results. However, if you're outputting an HDR deliverable, it's in your best interest to choose a wide gamut Color Space (and Gamma) to obtain the best results on output. In this instance, DaVinci Wide Gamut is a great choice (see below for more information).

No matter which Timeline Color Space you choose to work in, all clips in an edit are transformed from the Input Color Space that's either automatically or manually assigned to them, to the Timeline Color Space setting to provide the final output. This is how you can grade within a Log-encoded timeline color space and yet view a normalized or de-logged image.

IMPORTANT: Once you choose a Timeline Color Space and begin grading, do not change your Timeline Color Space, or you'll end up changing all of the grades that are built using the mathematics it defines. You can always change the Output Color Space to create a new deliverable, but all of your grades depend on the Timeline Color Space to render correctly.

DaVinci Wide Gamut Color Space and DaVinci Intermediate Gamma

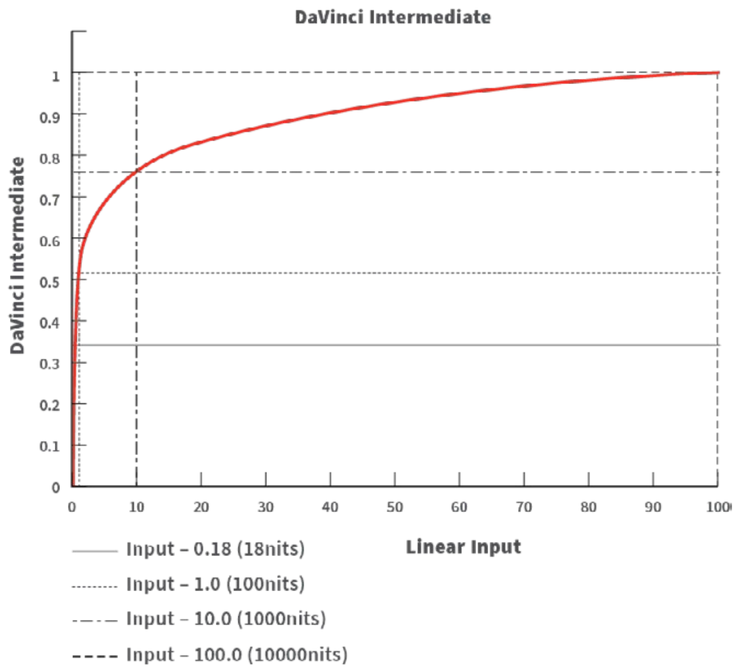
DaVinci Wide Gamut (DaVinci WG) and DaVinci Intermediate are Timeline Color Space and Gamma settings developed by Blackmagic Design that provide a reliable universal internal working color space, which encompasses a practical maximum of what image data any given camera can capture. The DaVinci Wide Gamut color space is greater than BT.2020, ARRI Wide Gamut, and even ACES, so you don't ever lose image data, no matter where your media is coming from.



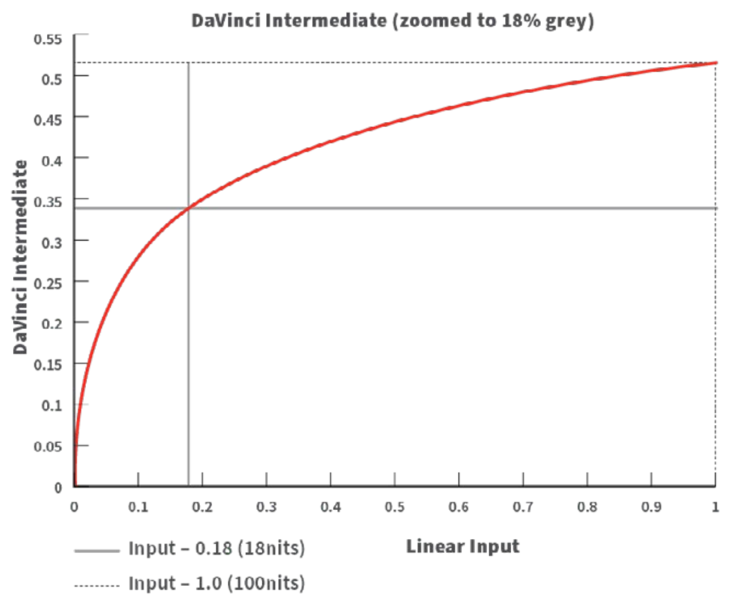
The DaVinci Wide Gamut color space

Furthermore, the primary color values of the DaVinci WG color space are set such that the process of automatically mapping source media from different cameras into this gamut is extremely accurate as part of the Input to Timeline Color Space conversion, and tone and saturation mapping from one color space to another can be done more accurately in the Timeline to Output Color Space conversion. This also helps to produce greater consistency among media from different cameras when making manual grading adjustments (though some variations due to differences in camera and lens systems will remain).

The DaVinci Intermediate OETF gamma setting has been designed to work with DaVinci Wide Gamut to provide a suitable internal luminance mapping of high precision image data, in preparation for mastering to either HDR or SDR standards, as your needs require, without losing image data.



The DaVinci Intermediate OETF seen encoding HDR levels



The DaVinci Intermediate OETF encoding SDR levels

Resolve Color Management is extremely flexible, so you don't have to use DaVinci Wide Gamut/DaVinci Intermediate as your Timeline color space if you don't want. However, it presents many advantages and is worth trying out to see if it can improve your workflow.

For more information, see the "DaVinci Resolve Wide Gamut Intermediate" document at https://www.blackmagicdesign.com/support/family/davinci-resolve-and-fusion.

Timeline Working Luminance

This control is only visible while the Resolve Color Management presets menu is set to Custom Settings. The Timeline Working Luminance drop-down menu lets you choose how the Input DRT (described below) maps the maximum level of a source image to the currently selected Timeline Color Space. This setting also defines the maximum highlight level that's possible to output into the currently selected Output Color Space using the Output DRT.

While it's typical to set this according to the mastering standard you're grading to via a collection of SDR and HDR labeled settings, there are additional settings available that make it possible to add more automatic compression of highlights as you grade.

SDR 100: The conventional setting for grading SDR material with a maximum level of 100 nits.

HDR 500-4000: Conventional settings for grading HDR material at a variety of maximum mastering levels. So long as output DRT isn't set to None, there will be some manner of rolloff in the highlights, unless inverse DRT is enabled, in which case there will be no rolloff.

SDR and HDR ER settings: These "extended range" settings each specify two values and provide more headroom for aggressive grading of highlights by enabling DaVinci Resolve to compress a greater range of out-of-bounds image data without clipping, which can result in a smoother look.

Here's how it works. Suppose you choose the setting "HDR ER 1000/2000." In this case, the Input DRT is used to map the maximum brightness of each source image to the range specified by the first value, which is 1000 nits. Then, when you grade, the signal isn't clipped until it reaches the maximum range specified by the second value, which is 2000 nits. This provides an additional 1000 nits of out-of-bounds headroom before the image data is hard clipped by RCM's image processing pipeline. The Output DRT is then used to map from the maximum brightness specified by the second number (2000 nits) to the output value defined by the currently selected Output Color Space, in the process compressing this out-of-bounds headroom to preserve as much highlight detail as is possible given the range you've selected.

Custom: Exposes a field where you can enter a specific nit value.

203 Nit Support for SDR to HDR

This control is only visible while the Resolve Color Management presets menu is set to Custom Settings. Resolve Color Management has support for remapping SDR content to HDR by mapping 100 nits to 203 nits (defined as the diffuse white level) according to the BT.2100 recommendation. This enables the peak highlights of SDR material to compete more favorably against the significantly brighter highlights of HDR content in programs that combine both (such as documentaries), so that SDR whites continue to appear white, rather than gray, when compared to diffuse white in HDR.

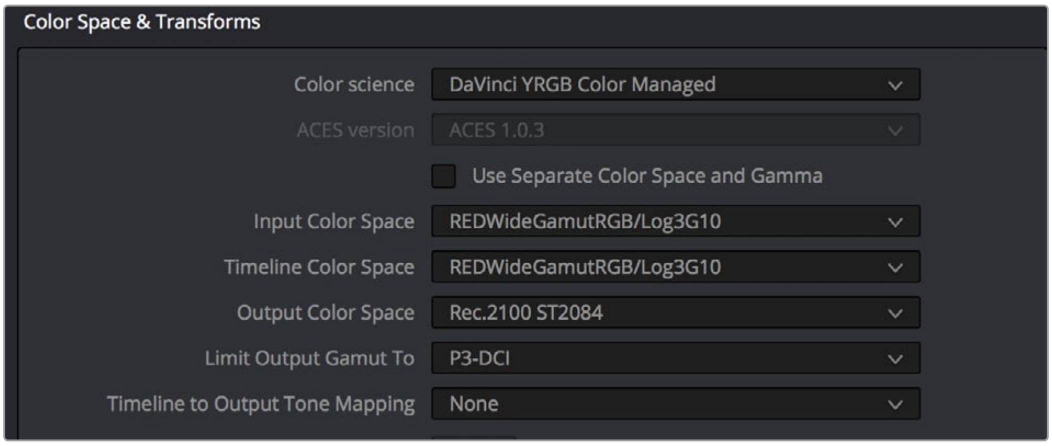
The checkbox that enables this is hidden by default. Whenever you set the Output to an HDR standard while the Timeline is set to an SDR standard, the “Use 203 nits reference for Rec.2100 HDR” checkbox for remapping SDR highlights to HDR appears in both the RCM settings of the Color Management panel of the Project Settings and in the Color Space Transform Resolve FX plugin.

The “Use 203 nits reference for Rec.2100 HDR” checkbox in Resolve Color Management for scaling SDR levels appropriately into HDR color space

Gamut Limiting, Restricting Values Within a Larger Gamut

This control is only visible while the Resolve Color Management presets menu is set to Custom Settings. In the emerging world of larger gamuts for distribution, it’s increasingly common for delivery specifications to specify output to a large gamut, such as Rec. 2020, yet require that image values be restricted to a smaller gamut, such as P3. This is to allow delivery to “future-proofed” delivery standards, while preventing saturation values that are too high to be displayed on consumer displays that aren’t capable of implementing the full scope of those standards.

In this case, you’ll choose a larger gamut in Output Color Space, but you’ll then choose a smaller gamut in “Limit Output Gamut To.” When you do this, all image values falling outside the “Limit Output Gamut To” standard specified will be hard clipped. This setting defaults to None.



Choose a setting from the Limit Output Gamut To menu to limit image values within a larger gamut

Input DRT Tone Mapping

This control is only visible while the Resolve Color Management presets menu is set to Custom Settings. RCM has always transformed the color primaries of different media formats to match one another within the shared Timeline Color Space. In this updated version, the Input DRT (Display Rendering Transform) drop-down menu provides a variety of different options to enable DaVinci Resolve to automatically tone map the image data of SDR and HDR clips to better match one another when they’re fit into the currently selected Timeline Color Space. While each option varies in the details, they are all automated input-to-timeline color transforms that do the following:

- Log-encoded media, or media using a 2.4 gamma transfer function, is mapped so the black point, midtones at 18% gray, and white levels match those of HDR media. Highlight data will be carefully stretched as necessary so that the highlights of all clips in the Timeline, whether SDR or HDR, are treated similarly.
- Raw formats such as BRAW, RED, and ARRI RAW, and media using HDR transfer functions are minimally mapped along an HDR range of tonality.
- All color transforms into the Timeline Color Space are done without clipping.

MEDIA

The idea is to distribute the image data of each clip in the Timeline, be it SDR or HDR media, along a similar histogram, with shadows, midtones, and highlights spread out in such a way as to create an easier starting point for grading. One result of this is that grades made for one type of media mostly work well with other types of media.

Different options are provided governing the details of how this Input to Timeline Color Space transform is achieved. They all do the same thing but have different advantages.

- **None:** This setting disables Input DRT Tone Mapping. No tone mapping is applied to the Input to Timeline Color Space conversion at all, resulting in a simple 1:1 mapping to the Timeline Color Space.
- **Simple:** A good mapping for color transforms from HDR to SDR.
- **Luminance Mapping:** Same as DaVinci, but more accurate when the Input Color Space of all your media is in a single standards-based color space, such as Rec. 709 or Rec. 2020.
- **DaVinci:** This option tone maps the transform with a smooth luminance roll-off in the shadows and highlights, and controlled desaturation of image values in the very brightest and darkest parts of the image. This setting is particularly useful for wide-gamut camera media and is a good setting to use when mixing media from different cameras.
- **Saturation Preserving:** This option has a smooth luminance roll-off in the shadows and highlights, but does so without desaturating dark shadows and bright highlights, so this is an effective option for colorists who like to push color harder. However, because over-saturation in the highlights of the image can look unnatural, two parameters are exposed to provide some user-adjustable automated desaturation.

  Sat. Rolloff Start: Lets you set a threshold, in nits (cd/m^2), at which saturation will roll off along with highlight luminance. Beginning of the rolloff.

  Sat. Rolloff Limit: Lets you set a threshold, in nits (cd/m^2), at which the image will be totally desaturated. End of the rolloff.

Output DRT Tone Mapping

This control is only visible while the Resolve Color Management presets menu is set to Custom Settings. To accommodate workflows where you need to transform one color space into another that has a dramatically larger or smaller gamut, an additional group of settings have been added that can help to automate the expansion or contraction of image data necessary to give a pleasing result.

Using the available options in the Output DRT drop-down menu will compress or expand your image data as necessary during the Timeline to Output Color Space transformation that RCM performs when monitoring or rendering a timeline, in order to make sure that the final result is either not clipping, or to ensure that it's taking better advantage of the new color space. This is not meant to provide your final grade. Rather, it's meant to give you a faster starting point, when you need it, for proceeding with your own more detailed grade of the result.

Here are some examples of what the Gamut Mapping controls of RCM can be used for:

1. If you're working with high-dynamic-range log-encoded media and you're outputting to Rec. 709 as you work, turning on Gamut Mapping lets RCM use saturation and tone mapping to give you a more immediately pleasing image with highlight detail that's not clipped.
2. If you're working with standard-dynamic-range log-encoded media and you're outputting to an HDR format as you work, turning on Gamut Mapping lets RCM use saturation and tone mapping to expand the highlights of the image to HDR strength to give you an image with more immediate visual impact on HDR screens.

(Before/After) Gamut Mapping used to automatically fit high-dynamic-range media into the Rec. 709 color space

The Output DRT (Display Rendering Transform) drop-down menu provides the following options.

- **None:** No tone mapping is applied to the Timeline to Output Color Space conversion at all, resulting in a simple 1:1 output with no softness or rolloff applied. All image data outside of gamut will be clipped.
- **Simple:** A good mapping for color transforms from HDR to SDR.
- **Luminance Mapping:** Same as DaVinci, but more accurate when all your media is in a single standards-based color space, such as Rec. 709 or Rec. 2020, set to the Timeline and Output.
- **DaVinci:** This option tone maps your output with a smooth luminance roll-off in the shadows and highlights, and controlled desaturation of image values in the very brightest and darkest parts of the image. It's been designed to give smooth, naturalistic highlights and shadows as you push and pull the values of your images, without the need for additional settings. This setting is particularly useful for wide-gamut camera media and is a good setting to use when mixing media from different cameras.

- **Saturation Preserving:** This option has a smooth luminance roll-off in the shadows and highlights to prevent clipping. It does so without desaturating dark shadows and bright highlights, so this is an effective option for colorists who like to push color a bit harder. However, because over saturation in the highlights of the image can look unnatural, two parameters are exposed to provide some user-adjustable automated desaturation.

  Sat. Rolloff Start: Lets you set a threshold, in nits (cd/m^2), at which saturation will roll off along with highlight luminance. Beginning of the rolloff.

  Sat. Rolloff Limit: Lets you set a threshold, in nits (cd/m^2), at which the image will be totally desaturated. End of the rolloff.

- **RED IPP2:** This setting lets you use RED IPP2 tone mapping to output to an SDR format, such as Rec. 709; two settings are exposed with which to choose how your output will be shaped.

  Output Tone Map: Lets you choose what kind of tone mapping you want to use for your output. Options include: None, Low, Medium, and High.

  Highlight Roll Off: Lets you choose what kind of highlight rolloff you want to use to prevent clipping. Options include: None, Hard, Medium, Soft, and Very Soft.

  HDR peak nits: A slider lets you choose the peak nit level you want to tone map to. Defaults to 10,000 nits.

Use Inverse DRT for SDR to HDR Conversion

This control is only visible while the Resolve Color Management presets menu is set to Custom Settings. A device rendering transform (DRT) is typically used when converting high dynamic range media to a lower dynamic range color space/mastering standard. Thus, setting up a color transform from SDR to HDR is an "inverse" operation to expand the dynamic range of SDR media to HDR standards. The way this works is that levels at 100 nits are mapped to the maximum value set for the Timeline Working Luminance parameter, and all other image levels are strategically tone mapped in order to give yourself a good starting point for grading SDR media into an HDR program.

This setting also has a secondary use. With this setting turned on, you can output Rec. 709 clips with color that's identical to the input, with no compression in the highlights.

NOTE: Turning on "Use Inverse DRT for SDR to HDR Conversion" may exaggerate noise in imported SDR media with large flat expanses of bright colors.

Use White Point Adaptation

This control applies a chromatic adaptation transform to account for different white points between color spaces.

- Uncheck this box if you simply want to view the input color space's white point unaltered in the output color space. For example, wanting to use a P3-D60 mastered clip inside a P3-D65 timeline for reference purposes.
- Check this box to apply the chromatic adaptation transform to convert the input white point to match the output color space's white point. For example, wanting a P3-D60 mastered clip to cut in with other clips mastered in a P3-D65 timeline.

NOTE: This control is only visible while the Resolve Color Management presets menu is set to Custom Settings.

Color Space Aware Grading Tools

In DaVinci Resolve version 17, both Resolve Color Management and ACES enables “color space aware” palettes, such as the new HDR palette, to have controls that feel consistent, no matter what color space the original media is from, or what Timeline Color Space you’re using.

Other palettes, such as the Qualifier and Curves palettes, become color space aware when you turn on the “Use Color Space Aware Grading Tools” checkbox in the Color Management panel of the Project Settings (this is turned on by default). When you’re using color space aware grading tools, you should not turn on HDR Mode for the node you’re working on.

- In the case of the Qualifier palette, this enables Qualifiers to create high-quality keys as you would expect, no matter what the color space of the original media is, or what Timeline Color Space you’re using.
- In the case of the Curves palette, this makes the overall range of each curve better fit the overall data range of the current clip, making curves adjustments easier and more specific.

NOTE: This control is only visible while the Resolve Color Management presets menu is set to Custom Settings.

Apply Resize Transformations In

When you’re using Resolve Color Management, a new “Apply Resize Transformations In” Project Setting is available in the Color Management panel while the Resolve Color Management presets menu is set to Custom Settings. This setting lets you choose which color space is used for resizing operations. Ordinarily, resizing is done in Linear, but certain specialty workflows benefit from doing resizing in other color spaces, so this option lets you choose which is best. The available options are:

Timeline: Uses the Timeline Color Space to perform all resizing operations.

Log: Uses a Log Color Space for resizing. Good for avoiding artifacts in certain high-contrast images, such as titles and star fields.

Linear: Usually provides the best results with most SDR media.

Linear Mapped: Usually provides the best results with most HDR media.

Gamma: Provided in case you find a need for this option.

Gamma Mapped: Usually provides best results when mixing SDR media with wide gamut and log-encoded media on the same timeline.

Graphics White Level

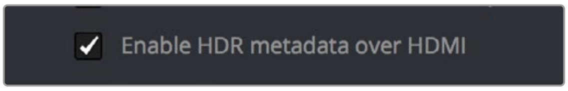
The “Graphics white level” setting lets you define a shared maximum level in nits (cd/m^2) for titles, generators, and selected effects that generate color. Changing this setting lets you change the maximum level of all DaVinci Resolve-generated titles, generator graphics, and effects at once to accommodate different mastering and output requirements.

Display HDR On Viewers If Available

Turn this checkbox on if your computer monitors and operating system are capable of accommodating HDR display. This allows the Viewers to show true HDR, according to the capabilities of your computer monitor.

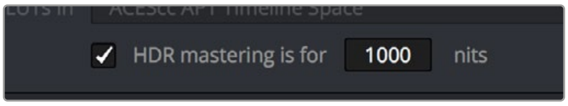
HDR Mastering Is For (Studio Version Only)

If you have a DeckLink 4K Extreme 12G or an UltraStudio 4K Extreme video interface, then DaVinci Resolve 12.5 and above can output the metadata necessary to correctly display HDR video signals to display devices using HDMI 2.0a when you turn on the “Enable HDR metadata over HDMI” checkbox in the Master Project Settings.



The Enable HDR metadata over HDMI option in the Master Project Settings lets you output HDR via HDMI 2.0a.

When you do so, a setting in the Color Management panel of the Project Settings, “HDR mastering is for X” lets you specify the output, in nits, to be inserted as metadata into the HDMI stream being output, so that the display you’re connecting to correctly interprets it. The output you specify should match what your display is expecting.



The “HDR mastering is for” setting lets you insert metadata for HDR output via HDMI 2.0a.

Resolve Color Management and the Fusion Page

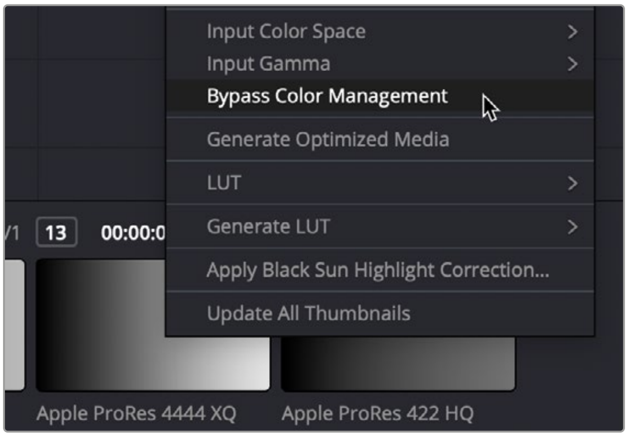
Enabling RCM also allows the Fusion page to handle the color of clips automatically. Images output by MediaIn nodes are automatically converted to Linear color space, which is the preferred color space with which to perform high-quality compositing operations. Setting the LUT menu of each Viewer in the Fusion page to Managed ensures that you’re looking at the image in Rec. 709, so that the image looks correct to the artist even though they’re really working in the Linear color space. Each MediaOut node then converts the image back to the timeline color space for handoff to the Color page.

With RCM off, you must manage color in the Fusion page manually, either using the Source Color Space and Source Gamma Space settings of each MediaIn node, or using the CineonLog or FileLUT nodes in your node tree.

For more information on how color management affects the Fusion page, and why the Linear color space is preferable for compositing, *see Chapter 76, “Controlling Image Processing and Resolution.”*

Ability to Bypass Color Management Per Clip

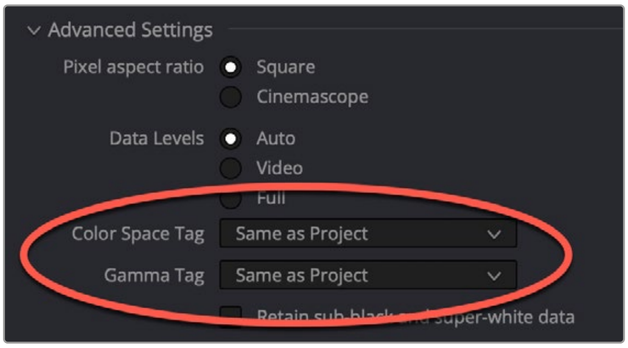
When you right-click a clip in the Thumbnail Timeline of the Color page, a “Bypass Color Management” setting appears underneath the Input Color Space and Input Gamma submenus that let you identify a clip’s color characteristics. Choosing this option so that it appears checked lets you exclude that clip from color management altogether, in the event you want to manually manage that clip using LUTs, the Color Space Transform node, or simply by doing manual grading.



The Bypass Color Management option for clips in the contextual menu of the Thumbnail Timeline

Exporting Color Space Information to QuickTime Files

If you render QuickTime files from the Deliver page, then color space tags will be embedded into each file based on either the Timeline Color Space (if Resolve Color Management is disabled) or the Output Color Space (if Resolve Color Management is enabled). Two settings in the Advanced Settings of the Render Settings let you choose how color space metadata will be embedded into your output for supported media formats, “Color Space Tag,” and “Gamma Tag.” These default to “Same as Project,” which will match the Output Color Space currently selected in the Project Settings.



The Color Space Tag and Gamma Tag settings in the Render Settings

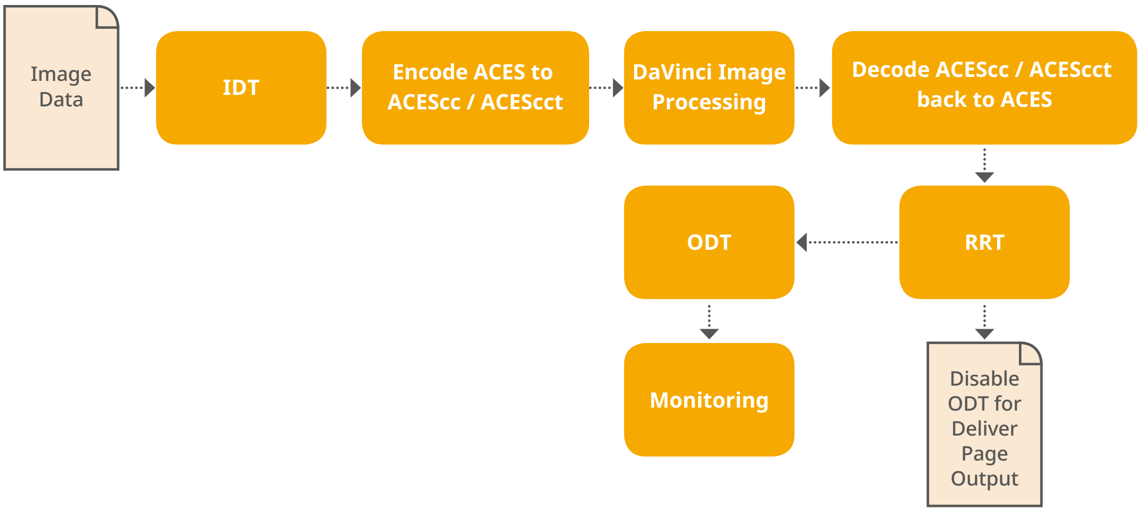
MEDIA

Color Management Using ACES

The ACES (Academy Color Encoding Specification) color space has been designed to make scene-referred color management a reality for high-end digital cinema workflows. ACES also makes it easier to extract high-precision, wide-latitude image data from raw camera formats, in order to preserve high-quality image data from acquisition through the color grading process, and to output high-quality data for broadcast viewing, film printing, or digital cinema encoding.

An oversimplification of the way ACES works is that every camera and acquisition device is characterized to create an IDT (Input Device Transform) that specifies how media from that device is converted into the ACES color space. The ACES gamut has been designed to be large enough to encompass all visible light, with more than 25 stops of exposure latitude. In this way ACES has been designed to be future-proof, taking into consideration advances in image capture and distribution.

Meanwhile, an RRT (Reference Rendering Transform) is used to transform the data provided by each image format's IDT into standardized, high-precision, wide-latitude image data that in turn is processed via an ODT (Output Device Transform). Different ODT settings correspond to each standard of monitoring and output, and describe how to accurately convert the data within the ACES color space into the gamut of that display in order to most accurately represent the image in every situation. The RRT and ODT always work together.



ACES signal and processing flow

By using the ACES color space and specifying an IDT and an ODT, you can ingest media from any capture device, grade it using a calibrated display, output it to any destination, and preserve the color fidelity of the graded image.

For more information on ACES see the ACES Central Website.

MEDIA

Setting Up ACES in the Project Settings Window

There are four parameters available in the Color Science drop-down of the Color Management panel of the Project Settings that let you set up DaVinci Resolve to use the ACES workflow:

Color science is: Using this drop-down menu, you can choose either DaVinci ACES, or DaVinci ACEScc color science, which enables ACES processing throughout DaVinci Resolve.

> **ACEScc:** Choose DaVinci ACEScc color science to apply a standard Cineon-style log encoding to the ACES data before it is processed by DaVinci Resolve. This well defined common encoding makes it possible for ASC CDL values to be used across systems using the same ACEScc encoding. After processing, a reverse encoding is applied in order to output ACES linear data.
>
> **ACEScct:** A variation of ACEScc that adds a roll-off at the toe of the image that's different from the encoding of ACEScc, in order to make color correction lift operations "feel" more like they do with film scans and LogC encoded images, which makes it easier to raise the darkest values of the image and get milky shadows, something that can be difficult with ACEScc. After processing, a reverse encoding is applied in order to output ACES linear data.

ACES Version: When you've chosen one of the ACES color science options, this drop-down becomes available to let you choose which version of ACES you want to use. You can choose from ACES 1.0.3, ACES 1.1, ACES 1.2, ACES 1.3, or ACES 2.0 (the latest version).

ACES AMF: For Aces 1.3 or above, you can select an ACES AMF file to be applied. More details can be found in the ACES AMF section below.

ACES Input Device Transform: This drop-down menu lets you choose which IDT (Input Device Transform) to use for the dominant media format in use. DaVinci Resolve currently supports the following IDTs:

> **ACEScc/ACEScct/ACEScg:** Standardized transforms for each of these ACES standards.
>
> **ADX (10 or 16):** 10-bit or 16-bit integer film-density encoding transforms meant for use if you're working with film scans that were initially encoded in an ACES workflow. This transform is designed to maintain the variation in look between different film stocks.
>
> **ALEXA:** Color management settings for all ARRI ALEXA cameras.
>
> **BMD Film/4K/4.6K:** Color management settings for Blackmagic Design cameras.
>
> **Canon 1D/5D/7D/C200/C300/C300MkII/C500/C700:** Color management settings for Canon cameras.
>
> **DCDM:** This IDT transforms X'Y'Z'-encoded media with a gamma of 2.6.
>
> **DCDM (P3D65 Limited):** This IDT transforms X'Y'Z'-encoded media with a gamma of 2.6, specifically hard clipped to a P3 gamut with a D65 white point.
>
> **DRAGONcolor/2 and REDgamma3/4/REDlogFilm combinations:** Different combinations of the DRAGONcolor, REDgamma, and REDlogFilm settings are provided for legacy RED workflows.
>
> **P3-D60:** Transforms RGB-encoded image data with a D60 white point, intended for monitoring with a P3-compatible display using a D60 white point.
>
> **P3-D65:** Transforms RGB-encoded image data with a D65 white point; intended for monitoring with a P3-compatible display using a D65 white point.
>
> **P3-D65 (D60 sim.):** Transforms RGB-encoded image data with a D65 white point; intended to simulate monitoring with a P3-compatible display using a D60 white point on a display with D65.
>
> **P3-D65 ST2084 (108/1000/2000/4000 nits):** Transforms an image that's compatible with the P3 color gamut, using the SMPTE standard PQ (ST.2084) tone curve for High Dynamic Range (HDR) post-production. Three settings for four different peak luminance ranges are provided; which one is appropriate to use depends on the maximum white level of the display used to create the media. Preliminary standards exist for HDR displays with peak luminance at 1000 nits, 2000 nits, and 4000 nits. A setting of 108 nits is provided for Kodak laser projection.

P3-D65: Transforms RGB-encoded image data with a D65 white point, intended for monitoring with a P3-compatible display using a D65 white point.

P3-D65 ST2084 (1000/2000/4000 nits): Transforms an image that's compatible with the P3 color gamut, using the SMPTE standard PQ (ST.2084) tone curve for High Dynamic Range (HDR) post-production. Three settings for three different peak luminance ranges are provided; which one is appropriate to use depends on the maximum white level of the display used to create the media. Preliminary standards exist for HDR displays with peak luminance at 1000 nits, 2000 nits, and 4000 nits.

P3-DCI (D60 sim.): Produces output that's specifically for output on a DCI projector with a D60 white point. This output may look magenta on other display devices that aren't set up for DCI display.

P3-DCI (D65 sim.): Produces output that's specifically for output on a DCI projector with a D65 white point. This output may look magenta on other display devices that aren't set up for DCI display.

Panasonic V35: Color management settings for each listed camera.

Rec.2020: This IDT transforms media created with the wide-gamut standard for consumer and broadcast television.

Rec.2020 ST2084 (1000/2000/4000 nits): This IDT transforms media created within the wide-gamut standard for consumer and broadcast television, using the SMPTE standard PQ (ST.2084) tone curve for High Dynamic Range (HDR) post-production. Three settings provided for HDR televisions with different peak luminance capabilities.

Rec.2020 HLG (1000 nits): This IDT transforms media within the wide-gamut standard for consumer and broadcast television and uses the Hybrid Log-Gamma (HLG) standard tone curve for High Dynamic Range (HDR) post-production. A single setting is provided for HDR televisions with peak luminance at 1000 nits.

Rec.709 (Camera): A deprecated legacy IDT for Rec. 709 that's included for backward compatibility. Converts the source data to linear based on Rec. 709 and transforms the result to ACES, but while this transformation is technically correct, it's not necessarily pleasing after conversion through the matching ODT. For this reason, the academy updated to the following Rec. 709 IDT, which is the inverse of the Rec. 709 ODT.

Rec.709: A standard transform designed to move media in the Rec. 709 color space into the ACES color space. This option is used for any other file type that might be imported, such as ProRes from Final Cut Pro, DNxHD from Media Composer, and any media file captured from tape.

Rec.709 (D60 sim.): A standard transform designed to move media in the Rec. 709 color space with a white point of D60 into the ACES color space.

REDColor2/3/4/REDGamma3/4/REDLogFilm combinations: Different combinations of the REDcolor, REDgamma, and REDlogFilm settings are provided for legacy RED workflows.

RWGLog3G10: The standardized RED IPP2 color pipeline transform for all RED camera media.

If you're working on a project that mixes media formats that require different IDTs, then you can assign different IDTs to clips using the Media Pool's contextual menu, or using the Clip Attributes window, which is also accessible via the Media Pool's contextual menu.

ACES Output Device Transform: This drop-down menu lets you choose an ODT (Output Device Transform) with which to transform the image data for monitoring on your calibrated display, and when exporting a timeline in the Deliver page. You can choose from the following options:

ADX (10 and 16): A standardized ODT designed for media destined for film output. Two settings accommodate 10-bit and 16-bit output. This ODT is not meant to be used for monitoring.

DCDM: This ODT exports X'Y'Z'-encoded media with a gamma of 2.6 intended for handoff to applications that will be re-encoding this data to create a DCP (Digital Cinema Package) for digital cinema distribution. This can be displayed via an XYZ-capable projector.

DCDM (P3D60 Limited): Outputs a P3 hard-limited signal with a D60 white point.

DCDM (P3D65 Limited): Outputs a P3 hard-limited signal with a D65 white point.

P3 D60: Outputs RGB-encoded image data with a D60 white point; intended for monitoring with a P3-compatible display using a D60 white point.

P3 D65: Outputs RGB-encoded image data with a D66 white point; intended for monitoring with a P3-compatible display using a D66 white point.

P3 D65 (D60 sim.): Outputs RGB-encoded image data to simulate monitoring with a P3-compatible display using a D60 white point on a display with a D65 white point.

P3 D65 (Rec.709 Limited): Outputs RGB-encoded image data with a D65 white point within a P3 gamut that's hard-limited to the color range of Rec. 709.

P3 D65 ST2084 (108/1000/2000/4000 nits): Outputs an image that's compatible with the P3 color gamut, using the SMPTE standard PQ tone curve for High Dynamic Range (HDR) post-production. Three settings for three different peak luminance ranges are provided; which one is appropriate to use depends on the maximum white level of your display. Preliminary standards exist for HDR displays with peak luminance at 1000 nits, 2000 nits, and 4000 nits. A setting of 108 nits is provided to simulate an HDR signal clipped to an SDR range.

P3 DCI (D60 sim.): Outputs RGB-encoded P3 image data that appears as if with a D60 white point on a DCI projector with a DCI white point.

P3 DCI (D65 sim.): Transforms RGB-encoded image data with a D61 white point (the DCI mastering standard) that appears as if with a D65 white point.

P3-D65 ST2084 (1000/2000/4000 nits): Transforms an image that's compatible with the P3 color gamut, using the SMPTE standard PQ (ST.2084) tone curve for High Dynamic Range (HDR) post-production. Three settings for three different peak luminance ranges are provided; which one is appropriate to use depends on the maximum white level of the display used to create the media. Preliminary standards exist for HDR displays with peak luminance at 1000 nits, 2000 nits, and 4000 nits.

Rec.2020: This ODT is for compatibility with the full range of this wide-gamut standard for consumer and broadcast television.

Rec.2020 (P3D65 Limited): Outputs a P3D65 hard-limited signal within this wide-gamut standard for consumer and broadcast television.

Rec.2020 (Rec.709 Limited): Outputs a Rec. 709 hard-limited signal within this wide-gamut standard for consumer and broadcast television.

Rec.2020 HLG: Outputs the full Rec. 2020 gamut to the Hybrid Log-Gamma standard for HDR.

Rec.2020 HLG (1000 nits, P3D65 Limited): Outputs a 1000 nit, P3D65 hard-limited signal within the Rec. 2020 gamut and the Hybrid Log-Gamma standard for HDR.

Rec.2020 ST2084 (1000/2000/4000 nits): This ODT transforms media created within the wide-gamut standard for consumer and broadcast television, using the SMPTE standard PQ (ST.2084) tone curve for High Dynamic Range (HDR) postproduction. Three settings are provided for HDR televisions with different peak luminance capabilities.

Rec.2020 ST2084 (1000/2000/4000 nits, P3D65 Limited): This ODT transforms media within the wide-gamut standard for consumer and broadcast television but with hard clipping at the boundary of the P3 gamut for televisions that are limited to the smaller P3 gamut for digital cinema; also uses the SMPTE standard PQ (ST.2084) tone curve for High Dynamic Range (HDR) post-production. Three settings are provided for HDR televisions with different peak luminance capabilities.

Rec.709: This ODT is used for standard monitoring and deliverables for TV.

Rec.709 (D60 Sim): A standard transform designed to move media in the Rec. 709 color space with a white point of D60 into the ACES color space.

sRGB: A standardized transform designed for media created for computer display in a consumer environment.

sRGB (D60 Sim): A standardized ODT designed for media destined for computer display in a consumer environment. Suitable for monitoring when grading programs destined for the web.

ACEScc/ACEScct/ACEScg: Standardized transforms for each of these ACES standards.

You must manually select an ODT that matches your workflow and room setup when working in ACES.

Process Node LUTs in: This drop-down menu lets you choose how you want to process CLF LUTs that are added to nodes in your grades while working in ACES, such as Look LUTs in on-set or VFX workflows. There are two choices: ACEScc AP1 Timeline Space (the default), and ACES AP0 Linear.

ACEScc AP1: For LUTs that have been designed to take the specific range of ACEScc data using the AP1 primary coordinates.

ACES AP0: For LUTs that have been designed for normal ACES data from 65504 to -65504 floating point values.

> **NOTE:** ACES grades require CLF LUTs that have been specifically created for ACES workflows. If you want to apply a regular LUT within a grade, you must do a color space transform to convert the image from ACES to whatever space the LUT was designed to work within, and then another color space transform to convert the image back to ACES; however, this workflow does not alway provide ideal results.

The Initial State of Clips When Working in ACES

Don't worry if the initial state of each image file appears differently than what was monitored originally on set. What's important is that if the camera original media was well exposed, the IDT used in ACES mode will retain the maximum amount of image data, and provide the maximum available latitude for grading, regardless of how the image initially appears on the Timeline.

ACES AMF 2.0

DaVinci Resolve supports ACES AMF 2.0 files for import and export. ACES AMF is a sidecar file written in XML that defines the input and output transforms of the clip. This makes sure that the correct input and output transforms are applied throughout the post production pipeline. These files can either be loaded at a project level or a clip level. Clips with the "applied" attribute set to true are tagged appropriately with no transforms applied. However, if they are set as false, the transform is applied within DaVinci Resolve.

> **NOTE:** You must use version 1.3 or above of the ACEScc or ACEScct Color Science to use AMF files.

Working with AMF Files

Project level AMF: These will have all transforms applied. (i.e., any defined input, look, or output transforms which have their "applied" attribute set to false will be applied.) The input and output transforms are applied via the project settings "ACES Input Transform" and "ACES Output Transform" menus, and these menus will display "From AMF" if they are using a transform from the loaded AMF file.

Clip level AMF: An AMF can be loaded from the Color page clip thumbnail context menu from the Clip AMF menu. This list is also populated from the ACES Transforms AMF subfolder. A clip-level AMF will not apply the output transform; that can only be applied from project settings. Any input or look transforms will have their "applied" attribute set to false. The look transforms are applied within a compound node labeled AMF LMTs.

ACES Gamut Compression: If the ACES reference gamut compress algorithm is specified in a look transform within the AMF, it will set the "Apply ACES reference gamut compress" checkbox to checked. The user can see what input, gamut compress, look and output transforms are applied by clicking the "View Transforms" button at the bottom of the "Color Space & Transforms" section of the Settings > Color Management section.

To Import an AMF

1. Place your AMF files in the ACES Transforms/AMF directory on your computer. This is folder is found:
   - **MacOS:** the users Library/Application Support/Blackmagic Design/DaVinci Resolve/Aces Transforms/AMF
   - **Windows:** the users AppData\Roaming\Blackmagic Design\DaVinci Resolve\Support\ACES Transforms\AMF
   - **Linux:** opt/resolve/Developer/DaVinciCTL/AcesTransforms/AMF
2. In the Project Settings, set the Color Management > Color Science setting to ACEScc or ACEScct.
3. Select the ACES version to be 1.3 or above.
4. Click on the ACES AMF drop-down menu, and select the AMF from the list.

Exporting AMF Files

There are three choices to select from when you export an AMF file.

Source Master mode:

- All the applied attributes will be set to false.
- A source master is not associated with any clips or images.

Dailies Request mode:

- You can select the folder to store the output AMFs.
- A dailies AMF is created for and associated with each clip, and the applied attribute will be set to false.

VFX Request mode:

- A VFX AMF will only have the input transform with the applied attribute set to false.

AMF Delivery Behaviors

- For AMF delivery, the export AMF option can be chosen in the Deliver page render settings.
- When using Source Master, the images are rendered to ACES AP0 or Linear masters with only the input transforms applied; all other transforms are set to applied false.
- The resulting AMF is not associated with any clips or images with an aim to define and transmit a color pipeline with its post production look alongside ACES AP0 or Linear data.
- In Dailies Request, all transforms are applied on the rendered images and all transforms are set to applied true.
- The resulting AMF is associated with each clip in the project.
- In VFX Request, the clips are rendered to ACES AP0 or Linear with only the input transform applied and all other transformed are set to false.

To Export an AMF

1. Right-click on your timeline in the Media Pool.
2. Select Timelines > Export > AMF in the contextual menu.
3. Choose the type of AMF file to export: Source Master, Dailies Request, or VFX Request.
4. Enter the name of the AMF file and its save location in the file browser.
5. Click on the Save button.

The Timeline Color Space in ACES Workflows is Fixed

When you're working in ACES, you do not get to change the Timeline Color Space as you do in Resolve Color Management. The ACES working color space is a log-encoded color space, which encourages a more traditional, film-oriented approach to grading.

Tips for Rendering Out of an ACES Project

When choosing an output format in the Deliver page, keep the following in mind:

- If you've delivering graded media for broadcast, set the ACES Output Device Transform to be Rec. 709, then you can output to whatever media format is convenient for your workflow.
- When you're delivering graded media files to another ACES-capable facility using the DCDM or ADX ODCs, you should choose the OpenEXR RGB Half (uncompressed) format in the Render Settings, and set the ACES Output Device Transform to "No Output Device Transform."
- When you're rendering media for long-term archival, you should choose the OpenEXR RGB Half (uncompressed) format in the Render Settings, and set the ACES Output Device Transform to "No Output Device Transform."